efficiency. Its stealth architecture includes radar-absorbent composite materials, internal weapon bays, and concealed air intakes that reduce radar visibility. The aircraft integrates AI-enabled avionics, sensor fusion, and digital flight control systems, enabling real-time situational awareness and autonomous decision support. Its modular mission systems allow role adaptability between air superiority, strike, electronic warfare (EW), and reconnaissance missions.

At the heart of AMCA lies an AI-driven avionics suite, comprising an integrated mission management system, data fusion algorithms, and a pilot-vehicle interface for manned operations. In its UCAV variant, these AI modules function autonomously, enabling the aircraft to perform missions such as target acquisition, threat evaluation, and evasive maneuvers without pilot input. Additionally, the AMCA incorporates network-centric warfare capabilities, allowing seamless data exchange with drones, satellites, and ground-based assets for cooperative engagement (Yelwe, 2025).

Powered initially by GE F414 engines (with plans for an indigenous joint Indo-French engine in later variants), the AMCA is expected to achieve super cruise speeds exceeding Mach 2 and an operational range of over 2000 km. Its dual-seat configuration variant is planned for advanced mission roles and loyal-wingman control, pairing with future autonomous UAVs under the Combat Air Teaming System (CATS) initiative.

Strategically, the AMCA symbolizes India's entry into the elite group of nations capable of developing fifth-generation stealth combat aircraft. It not only strengthens the country's defense deterrence and export potential but also lays the technological groundwork for future sixth-generation adaptive combat platforms.

Figure 6.1 depicts India's Advanced Medium Combat Aircraft (AMCA), a next-generation fifth-generation multirole stealth fighter designed by the